

JE Civil — Final Deep Explanations

ಅಂತಿಮ ಆಳವಾದ ವಿವರಣೆ — ಎಲ್ಲಾ ವಿಷಯಗಳು

KRCL CO/P-R/02/2026 · English + ಕನ್ನಡ · Teacher-style topic notes

Science · Maths · GA · Reasoning · Materials · Survey · Structures · Fluids · Geotech/Transport ·
Env/Estimation

Final Deep Explanations — All Topics

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KRCL JE Civil (CO/P-R/02/2026) · English + ಕನ್ನಡ · Teacher-style deep notes

How to use: Read one topic fully → solve 10 MCQs from that topic → revise “Must lock” box.

1. General Science | ಸಾಮಾನ್ಯ ವಿಜ್ಞಾನ

What exam expects | ಪರೀಕ್ಷೆ ಏನು ಕೇಳುತ್ತದೆ

Class 10 CBSE basics: motion, force, energy, light, electricity, acids-bases, metals, life processes, environment. Not engineering physics — **definitions + units + simple relations.**

Deep explanation (Physics)

Force, work, energy — one story

Imagine pushing a box:

- **Force** makes velocity change $\rightarrow F = ma$

ಬಲ = ದ್ರವ್ಯರಾಶಿ $\times$ ವೇಗವರ್ಧನೆ

- If the box **moves** in force direction $\rightarrow$ **Work** $W = F \times s$

ಚಲನೆಯ ದಿಕ್ಕಿನಲ್ಲಿ ಬಲ $\rightarrow$ ಕೆಲಸ

- If force is **perpendicular** to motion (like circular string tension sideways to instantaneous displacement idea in basics) $\rightarrow$ work can be zero in school definition when displacement $\perp$ force.
- Energy is "ability to do work": moving $\rightarrow KE = \frac{1}{2}mv^2$; height $\rightarrow PE = mgh$.

Teacher tip: Same joule unit for work and energy because energy is stored work capacity.

Electricity chain

Current I is flow of charge. Battery gives voltage V . Resistance R opposes.

- Ohm: $V = IR$
- Series: resistances add (one path)
- Parallel: reciprocals add (multiple paths $\rightarrow$ less total R)

Real feel: Festival lights in series — one bulb fails, all die. House wiring parallel — one bulb off, others work.

Light & sound

- Light: reflection $i = r$; convex converges, concave diverges.
- Sound needs medium; vacuum silent. Speed in air ~ 340 m/s.

Deep explanation (Chemistry)

- $pH < 7$ acid, $= 7$ neutral, > 7 base. Litmus: acid turns blue $\rightarrow$ red.
- Rusting = $Fe + O_2 + \text{moisture}$. Prevent by paint/oil/galvanizing.
- Brass = $Cu+Zn$; Bronze = $Cu+Sn$.

Deep explanation (Biology/Env)

- Photosynthesis in chloroplast: $CO_2 + H_2O \rightarrow \text{glucose} + O_2$ (sunlight).
- Heart 4 chambers; kidneys purify; ozone blocks UV; CO_2 is greenhouse gas.

Must lock | ಅನಿವಾರ್ಯ

$F=ma$, $W=Fs$, $KE=\frac{1}{2}mv^2$, $PE=mgh$, $V=IR$, pH7 water, photosynthesis products, rust needs air+moisture.

Exam traps

- Confusing power (Watt) with energy (Joule).
 - Saying sound travels in vacuum.
 - Mixing brass and bronze.
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2. Basic Mathematics | ಮೂಲಭೂತ ಗಣಿತ

What exam expects

Speed calculation under pressure. 20 Q $\times$ 3 marks = 60 marks. Accuracy > attempting all with guessing (-1).

Deep explanation — how to think

Percentages are fractions in disguise

10% = $1/10$, 25% = $1/4$, 12.5% = $1/8$. Increase $x\%$ $\rightarrow$ multiply by $(1 + x/100)$. Successive $a\%$ then $b\%$ $\neq$ $(a+b)\%$ exactly; extra $ab/100$ term.

Time & Work is “rate”

If A finishes in a days, rate = $1/a$ per day. Together: add rates. Pipes: inlet +, outlet -.

Time & Distance

$D = S \times T$. Always match units. $\text{km/h} \rightarrow \text{m/s} \times 5/18$; reverse $\times 18/5$. Equal distances, speeds $x, y \rightarrow \text{avg} = \frac{2xy}{x+y}$.

SI vs CI

SI grows **linear** with time. CI grows on previous amount — compound. For 2 years, $\text{CI} - \text{SI} = P(R/100)^2$ (standard result).

Geometry / Trig essentials

Triangle angles 180° . Pythagoras $a^2+b^2=c^2$. $\sin=\text{opp}/\text{hyp}$, $\cos=\text{adj}/\text{hyp}$, $\tan=\text{opp}/\text{adj}$. $\sin^2+\cos^2=1$. Memorize $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$ table.

Worked mini examples

- 15% of 200 = 30.
- $36 \text{ km/h} = 36 \times 5/18 = 10 \text{ m/s}$.
- $A=10\text{d}, B=15\text{d} \rightarrow$ together $1/10+1/15=1/6 \rightarrow 6$ days.
- SS not needed here — keep rates clear.

Must lock

BODMAS, $\text{HCF} \times \text{LCM} = \text{product}$ (2 numbers), $\text{SI} = \text{PRT}/100$, $5/18$ and $18/5$, $\text{WL}/4$ and $\text{WL}^2/8$ appear in SOM too, $\tan 45=1$.

Exam traps

Doing left-to-right ignoring BODMAS. Using SI formula for CI. Forgetting unit conversion in speed.

3. General Awareness | ಸಾಮಾನ್ಯ ಜ್ಞಾನ

Deep explanation — how to remember India

Build 5 pillars:

1. **Polity:** Constitution 26 Jan 1950; Ambedkar drafting; FR Part III; Art 14,19,21,32.
2. **Geography:** Himalaya–plains–Deccan; Ganga longest among majors often asked; monsoon; black soil→cotton.
3. **History:** 1857, INC 1885, Non-cooperation 1920, Civil Disobedience 1930, Quit India 1942, Freedom 15 Aug 1947.
4. **Economy:** RBI central bank; GDP/inflation idea; GST; primary/secondary/tertiary.
5. **Env/Sports/Tech:** Project Tiger, Paris Agreement, ISRO Chandrayaan, Arjuna award.

Karnataka anchors: Bengaluru capital; Kaveri; Western Ghats; Bandipur; Hampi (Vijayanagara).

Must lock

26 Jan 1950, Ambedkar, Art 21, Quit India 1942, RBI, GST meaning, BG not needed here.

Exam traps

Confusing EWS (job reservation) with EBC (fee only in many exams). Mixing Fundamental Rights and DPSP parts.

4. Reasoning | ਤਰਕਸ਼ਕਤੀ

Deep explanation — pattern hunting

Every reasoning Q is “find the rule”:

- Series: +2, $\times 2$, squares, primes, alternate patterns.
- Coding: +1 letter, reverse alphabet, position sums.
- Blood relation: draw family tree; don't solve in head.
- Direction: sketch N/E/S/W; right/left turns.
- Syllogism: Venn; “all/some/no”; don't assume outside info.
- Statement-assumption: hidden belief needed for statement to make sense.

Must lock

A=1...Z=26; BB style not needed; draw diagrams always.

Exam traps

Using outside knowledge in syllogism. Mixing left/right when facing south.

5. Building Materials & Construction | ಕಟ್ಟಡ ಸಾಮಗ್ರಿ

Deep explanation — materials as a system

Building stands because materials carry load and resist weather.

Bricks

Good brick: uniform, ringing sound, low water absorption, strength $\geq 10.5 \text{ N/mm}^2$ (1st class). Underburnt = soft/weak; overburnt (jhama) = distorted, sometimes used in foundations as hardcore idea.

Cement chemistry (exam gold)

- **C3S**: early strength (first days).
- **C2S**: later strength.
- **C3A**: high heat of hydration.

Initial set $\geq 30 \text{ min}$ (working time); final $\leq 10 \text{ hr}$.

Water-cement ratio (Abrams)

Less water (enough for hydration/workability) $\rightarrow$ denser paste $\rightarrow$ **higher strength**. Too little water $\rightarrow$ poor compaction; too much $\rightarrow$ pores $\rightarrow$ weak.

Workability

Slump test measures ease of placing. Heavily reinforced needs higher slump.

Concrete age

Standard cube 150 mm; 28-day strength is characteristic reference.

DPC

Moisture rises by capillary action into walls $\rightarrow$ damp. DPC at plinth stops rising damp.

Must lock

10.5 N/mm^2 , initial set $\geq 30 \text{ min}$, lower w/c $\rightarrow$ higher strength, slump=workability, DPC at plinth, M20 $\approx$ 1:1.5:3, M15 $\approx$ 1:2:4.

Exam traps

Higher w/c means higher strength (false). Final setting 30 min (false — that's initial minimum).

6. Surveying & Levelling | ಸಮೀಕ್ಷೆ

Deep explanation — from whole to part

You first establish accurate control (big framework), then detail. Prevents error accumulation.

Compass bearings

WCB $0-360^\circ$. Back bearing = Fore $\pm 180^\circ$. If $FB-BB \neq 180^\circ \rightarrow$ local attraction (iron/current disturbing needle).

Levelling arithmetic

$HI = RL + BS$ Next $RL = HI - FS$ (or IS) This is the entire book-keeping of height transfer.

Contours

Equal elevation lines. Close spacing = steep. Closed with higher inside = hill.

Modern tools

Theodolite: angles. Tachometry: distance from staff intercept ($D \approx 100s + C$). Total Station: angles+distance+coords. GPS: satellites.

Must lock

Whole $\rightarrow$ part, $BB = FB \pm 180^\circ$, $HI = RL + BS$, close contours = steep, $K \approx 100$.

Exam traps

$HI = RL - BS$. Thinking widely spaced contours are steep.

7. Mechanics & Structural | ರಚನೆ

Deep explanation — stress to structure

Stress & strain

Stress = load intensity (P/A). Strain = stretch ratio ($\delta L/L$). Within elastic limit, Hooke: stress $\propto$ strain $\rightarrow E = \sigma/\epsilon$.

SFD & BMD (why engineers draw them)

Beam has shear and bending. Failure often starts where BM or shear is critical. Relations: $dM/dx = SF$ (sign convention dependent). Max BM often where $SF=0$.

Lock formulas:

- SS, central point load: $M_{max} = WL/4$
- SS, UDL w : $M_{max} = wL^2/8$

RCC philosophy

Concrete strong in compression, weak in tension $\rightarrow$ steel in tension zone. **Under-reinforced:** steel yields first $\rightarrow$ ductile warning $\rightarrow$ preferred. **Over-reinforced:** concrete crushes suddenly $\rightarrow$ brittle $\rightarrow$ avoid. Stirrups fight shear near supports.

Slabs

$L_y/L_x > 2 \rightarrow$ one-way (main steel short span). $\leq 2 \rightarrow$ two-way.

Steel

Compression members buckle; capacity falls with high slenderness L_e/r . Fillet weld throat $\approx 0.7 \times$ size.

Must lock

$\sigma = P/A$, $WL/4$, $wL^2/8$, under-reinforced preferred, $L_y/L_x > 2$ one-way, throat $\approx 0.7t$, Fe415 $f_y \approx 415$ MPa.

Exam traps

Preferring over-reinforced. Mixing SF max location with BM max blindly without conditions.

8. Fluid Mechanics & Hydraulics | ದ್ರವಗಳು

Deep explanation — pressure and energy

Pressure in liquid

$P = \rho gh$. Deeper $\rightarrow$ more pressure (dam bottom thicker conceptually).

Continuity

Mass conserved. For incompressible: $A_1V_1 = A_2V_2$. Narrow pipe $\rightarrow$ faster flow.

Bernoulli (energy per unit weight)

$P/\gamma + V^2/2g + z = \text{constant along streamline (ideal)}$. Venturi, orifice, pitot use this idea.

Pipe losses

Friction dominates long pipes: $h_f = fLV^2/(2gD)$. Reynolds number: low $\rightarrow$ laminar ($\sim < 2000$), high $\rightarrow$ turbulent ($\sim > 4000$).

Hydrology words

Rain gauge measures rainfall. Run-off = rain that flows on surface. Duty/delta relate irrigation supply to crop water.

Must lock

$P = \rho gh$, $A_1V_1 = A_2V_2$, Bernoulli three heads, Re limits, rain gauge.

Exam traps

Bernoulli without knowing each term meaning. Thinking continuity increases both A and V together.

9. Geotech & Transportation | ಮಣ್ಣು / ಸಾರಿಗೆ

Deep explanation — soil decides foundation

Structure load eventually sits on soil. Soft clay → settlement/consolidation with time. Sand drains faster.

Index properties

Water content w , void ratio e , porosity n , saturation S . Important: $Se = wG$.

Atterberg

LL: starts flowing. PL: thread crumbles. $PI = LL - PL$ (clayey if high).

Compaction

Beat out air voids. OMC → MDD peak on curve. More energy → higher MDD, lower OMC typically.

Foundations

Shallow (footing/raft) if good soil near surface. Deep (piles) if weak upper layers. Differential settlement damages buildings more than uniform.

Highways

Camber drains water. SSD for stopping. Flexible = bituminous layers; rigid = CC slab.

Railway

BG = 1.676 m. Sleepers hold gauge; ballast drains + cushions.

Must lock

$Se = wG$, $PI = LL - PL$, OMC/MDD, SSD idea, BG = 1.676 m, flexible vs rigid.

Exam traps

MG and BG mix-up. Thinking rigid pavement is bituminous.

10. Environmental Engineering & Estimation | ಪರಿಸರ / ಅಂದಾಜು

Deep explanation — water to waste to cost

Water treatment story (sequence matters)

Raw water → screening → sedimentation → coagulation/filtration ideas → **filtration** → **chlorination** → distribution. Residual chlorine protects in pipes.

Sewage

BOD = oxygen microbes need for organics (pollution indicator). Primary = physical settling/screening. Secondary = biological (ASP, trickling filter).

Estimation

Detailed estimate = quantity × rate from drawings. Rate analysis = material + labour + equipment + overheads + profit. Tender pack: NIT + specs + BOQ + conditions. EMD shows seriousness.

Must lock

Sedimentation→Filtration→Chlorination, high BOD=more pollution, primary vs secondary, rate analysis components, BOQ/NIT.

Exam traps

Wrong treatment order. Thinking BOD measures hardness.

Cross-topic exam strategy | ಒಟ್ಟು ತಂತ್ರ

1. Technical 50Q = 150 marks → master topics 5–10 first.
2. Maths 20Q = fast marks if formulas fluent.
3. Never random-guess 4 options (−1 expected value ~0). Eliminate to 2, then attempt.
4. Weekly: 1 Full Mock + error notebook.
5. Last 7 days: formula sheet + deep “Must lock” only.

Target practice score: 210+ / 300.

End of Final Deep Explanations · KRCL JE Civil